

Lateral Navigation in the Browser Boosts Health Misinformation Detection Even Under Confirmation Bias

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Abstract

Health misinformation poses serious public health risks, yet little is known about how ordinary users verify such content in real time. In this study, we investigate how behavioral fact-checking strategies, specifically, lateral reading (consulting multiple external sources) versus vertical reading (spending more time on fewer sources), affect accuracy in detecting health-related misinformation. We conducted a large-scale online experiment with 1,842 participants using a designed for the experiment purposes standalone web platform that logs respondents' behavioral signals (tab/window switches and time spent outside the experimental interface). This design enables indirect measurement of fact-checking in an ecologically valid setting.

Using linear mixed-effects models with semiparametric bootstrap confidence intervals, we find that lateral reading significantly improves accuracy ($\beta = 0.09$, 95% CI [0.079, 0.181], $p = 0.001$), while vertical reading has no effect ($\beta = -0.01$, $p = 0.332$). Crucially, confirmation bias, operationalized as alignment between statement valence and user attitude, does not moderate this relationship ($p = 0.876$). This suggests that lateral reading remains effective even when users encounter ideologically congruent misinformation.

Our results demonstrate that simple, scalable behavioral cues like tab-switch frequency can serve as reliable indicators of verification quality. For web platforms and digital literacy initiatives, this implies that nudging users to “open another tab” may be a lightweight, bias-resistant intervention to improve misinformation resilience, especially in health contexts where errors carry real-world consequences.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI) → Empirical studies in HCI • Applied computing → Law, social and behavioral sciences → Psychology • Information systems → World Wide Web → Web mining → Web content analysis

Keywords

misinformation detection, fact-checking, lateral reading, behavioral logging, confirmation bias, web experiment, digital literacy, health communication

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1 Introduction

The web has transformed ordinary users into active agents of information creation and verification. Yet this empowerment comes with a serious risk: health-related misinformation spreads faster and wider than factual content [1] and can delay medical treatment, reduce vaccine uptake, and even increase mortality [2, 3]. While journalistic and automated fact-checking are well studied [4, 5], health information verification by ordinary users remains poorly understood, especially the natural fact-checking strategies they employ and how these strategies affect detection accuracy.

Fact-checking is not merely a cognitive judgment; it is a browser-native behavior. Users may stay on the original page for a long time (vertical reading) or open several new tabs to consult other sources (lateral reading). These actions generate passive, scalable digital footprint as tab switches and time away from the interface,



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that can be logged directly in the browser without extensions or surveys [6, 7]. Prior work shows lateral reading improves credibility assessment among trained individuals [8], but its effectiveness among untrained users in real-world settings is unclear.

RQ1: *Do naturally occurring fact-checking strategies, specifically, (1) lateral and (2) vertical reading, differentially predict accuracy in health misinformation detection?*

Even when users engage in fact-checking, their judgments and web searches may be undermined by confirmation bias (CB), the tendency to accept information that aligns with pre-existing beliefs and dismiss contradictory evidence [9]. This bias reduces critical scrutiny, especially for ideologically congruent health information, and can render fact-checking ineffective [10, 11]. If confirmation bias weakens the link between fact-checking and accuracy, then behavioral interventions may fail precisely when they are most needed.

RQ2: *Is the relationship between these behavioral strategies and accuracy moderated by confirmation bias (i.e., alignment between user attitude and claim’s valence)?*

Our study leverages 64,020 behavioral observations from 1,842 participants on a custom-built web platform that logs tab-switches and time outside the interface in real time, offering a rare, ecologically valid window into how verification unfolds in a high-stakes domain.

2. Related Work

2.1 Fact-checking beyond journalism

Traditional research frames fact-checking as a journalistic practice [4, 12]. However, in the digital era, ordinary users have become active participants in information verification [13]. While recent studies show that lateral reading, consulting multiple external sources, improves credibility assessment among trained users [7, 8], evidence remains scarce for untrained individuals in real-world web environments.

2.2 Web-native behavioral measurement.

Most existing work relies on self-reports or lab-based eye-tracking, which suffer from low ecological validity and poor scalability [14]. In contrast, passive logging of native browser events, such as tab-switching, enables unsupervised, large-scale measurement of verification behavior as it naturally occurs in the browser.

2.3 Confirmation bias and misinformation

CB reduces critical scrutiny and increases vulnerability to health misinformation [9, 11]. Crucially, it remains unknown whether behavioral fact-checking strategies like lateral reading can overcome this bias in ecologically valid settings.

Our work bridges these gaps. We measure real-time, browser-native fact-checking behavior among 1,842 untrained users during health misinformation assessment. We test whether different fact

checking strategies improve detection accuracy—and, critically, whether this effect persists under confirmation bias.

3 Method

Participants completed the study on a standalone, custom-built web platform (<https://exp.linis-crowd.org/shared>) that required no browser extensions or client-side installations. The interface presented 30 health-related statements one at a time in random order. For each statement, participants rated its truthfulness on a 6-point scale (1 = definitely false to 6 = definitely true). They were explicitly informed they could consult external resources and were subtly encouraged to prioritize accuracy – mimicking real-world verification behavior.

3.1 Material

The stimulus set consisted of 56 verified health claims (28 true, 28 false) covering five domains: vaccines, GMOs, vitamins, alternative medicine, and lifestyle.

True statements were grounded in evidence from authoritative sources (e.g., WHO, Cochrane reviews); false statements were factually incorrect yet plausible, adapted from real-world health myths circulating in Russian-language online spaces (e.g., social media, forums, and popular blogs).

All items **underwent a rigorous multi-stage validation process** conducted by three medical professionals (MDs with clinical experience) and a senior meta-reviewer. A statement was assigned its final label only after consensus among annotators, ensuring high stimulus validity and ecological relevance to real-world fact-checking challenges (see Figure 1).

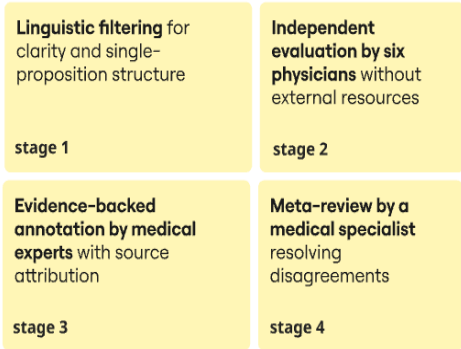


Figure 1: **Multi-Stage Expert Validation Process for the Dataset**

3.2 Measurements

Following Wineburg & McGrew [6], we distinguish two fact-checking strategies based on browser behavior:

- Lateral reading characterized by frequent tab/window switches, reflecting brief consultation of multiple external sources;

Table 1: Mixed-Effects Model Predicting Perceptual Accuracy Output: Fixed Effects Predicting Accuracy

Predictor	β	95% CI (bootstrap)	p
Fact-checking behaviors			
Tab-switches (lateral reading)	0.09	[0.079, 0.181]	0.001
Time away (vertical reading)	-0.01	[-0.038, 0.007]	0.332
Moderation by confirmation bias			
Tab-switches $\times$ Consistency	-0.01	[-0.070, 0.074]	0.876
Time away $\times$ Consistency	-0.03	[-0.071, 0.009]	0.165
Significant controls			
Male (vs. Female)	-0.11	[-0.135, -0.071]	<0.001
Education (only for Secondary Professional)	-0.12	[-0.165, -0.079]	<0.001
Political conspiracy belief	-0.15	[-0.187, -0.116]	<0.001
Religious conspiracy belief	-0.11	[-0.145, -0.076]	<0.001
Subjective expertise	0.05	[0.040, 0.064]	<0.001

Bold for the significant effect

- Vertical reading characterized by prolonged time spent outside the experiment interface, reflecting deep engagement with one or few sources.

While the ideal lateral reader exhibits high tab-switching frequency and low time per tab, and the ideal vertical reader shows low switching frequency and high time per tab, our design treats these dimensions independently to assess the contribution of each behavioral signal to accuracy.

Confirmation bias was operationalized as a binary variable (consistency) indicating alignment between:

- Participant attitude (e.g., “Would you get vaccinated if recommended by a doctor?”), and
- Statement valence (positive/negative framing of the topic).

Accuracy measured as 1–6 scale (1 = completely incorrect, 6 = completely correct), aligned with ground-truth veracity.

Control variables:

- Gender, age, education level;
- Conspiracy beliefs (political, religious);
- Text topic (Vaccines, GMOs, Vitamins, Lifestyle, Alternative Medicine).

All behavioral data were collected between November 2023 and February 2024 via Yandex Toloka (N = 1,842; M age = 40.7, SD = 12.4). The final dataset contains 64,020 observations.

4 Analysis and Result

Statistical analysis employed linear mixed-effects models (LMMs) predicting accuracy (ACC), with random intercepts for participants and texts.

Although accuracy was measured on a 6-point ordinal scale, we modeled it as a continuous outcome using linear mixed-effects regression. This approach is widely adopted in research when the ordinal scale has ≥ 5 points and approximates an interval-like structure [15, 16]. Diagnostic

checks confirmed that residuals were approximately symmetric and homoscedastic at the group level, and the large sample size (N = 64,020 observations across 1,842 participants) ensured robustness [16]. We opted for LMM due to its interpretability, computational stability, and compatibility with bootstrap-based inference (lme4::bootMer).

The model included random intercepts for both participants and statements to account for repeated measures and item-level variability. Fixed effects comprised lateral reading (tab-switches), vertical reading (time away), their interaction with confirmation bias (consistency), and control variables (age, gender, education, conspiracy beliefs, and topic). Confidence intervals were estimated using semiparametric bootstrap (1,000 replications) via lme4::bootMer in R to relax distributional assumptions.

4.1 Lateral reading improves accuracy

Each additional tab switch increased accuracy by 0.09 points (95% CI [0.079, 0.181], $p = 0.001$). In contrast, time spent outside the tab had no effect ($\beta = -0.01$, $p = 0.332$). This supports RQ1: breadth > depth in fact-checking.

4.2 Confirmation bias does not moderate the effect

The interaction between tab-switching and confirmation bias (consistence) was non-significant ($\beta = -0.01$, $p = 0.876$). Similarly, time $\times$ consistency showed no moderation ($p = 0.165$). Thus, lateral reading worked equally well for pro- and counter-attitudinal statements, answering RQ2.

4.3 Control variables

Higher accuracy was linked to subjective expertise ($\beta = 0.05$, $p < 0.001$) and lower conspiracy mentality (political: $\beta = -0.15$; religious: $\beta = -0.11$; both $p < 0.001$).

5 Discussion and Implications

Our results demonstrate that tab-switch frequency, a lightweight, client-side behavioral signal derived from native

browser events, robustly predicts improved health misinformation detection, even among non-expert users.

Crucially, this effect holds under confirmation bias, suggesting that consulting multiple sources inherently mitigates selective exposure by exposing users to diverse informational contexts.

The null result for time away should be interpreted not as evidence against vertical reading per se, but as a limitation of available behavioral signals in extension-free web environments. Prolonged absence from the interface may reflect distraction or multitasking rather than deep engagement – a known issue with time-based metrics in unsupervised settings [14]. This underscores the challenge of measuring depth without compromising privacy or ecological validity.

These findings point to a simple, technically feasible intervention: nudges that encourage lateral navigation, such as “Open another tab to verify.” Compared to generic prompts like “read carefully,” such cues are:

- **Lightweight:** no need for AI, fact-checking databases, or user training,
- **Scalable:** based on universally available browser signals (blur/focus),
- **Bias-resistant:** effective even for ideologically congruent misinformation.

Such interventions could be deployed during health-related search sessions (e.g., on Google, Yandex) via temporary UI banners or search result annotations.

While our behavioral proxies – tab-switch frequency and time away – are intentionally coarse, they were chosen to maximize ecological validity and scalability without requiring browser extensions, eye-tracking, or intrusive instrumentation. Richer signals (e.g., scroll depth, semantic analysis of visited sources) remain inaccessible in standard web environments due to privacy constraints. Future lab-based studies could triangulate our findings with multi-modal data (e.g., eye-tracking + tab logs), but for large-scale deployment, tab-switching remains a uniquely practical indicator of effective fact-checking.

6 Conclusion

We demonstrate that lateral navigation in the browser – a behavior as simple as switching tabs – is a potent, bias-resistant predictor of health misinformation detection accuracy. By leveraging native web signals, our approach offers a scalable path toward embedding digital literacy directly into the fabric of online interaction. For The Web Conference community, this underscores the value of behavioral observability: sometimes, the most powerful interventions start with measuring what users already do.

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